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A Selection-Bounded Mass–Metallicity Deficit at $z > 7$

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ABSTRACT

We report a *bounded, automated, descriptive result — not a validated measurement* — on the gas-phase mass–metallicity relation (MZR) at $z > 7$. Placing an SDSS local anchor ($N = 203{,}599$ star-forming galaxies, Tremonti et al. 2004 [T04] scale) and the JWST spectroscopic census of Nakajima et al. (2023) onto a single abundance scale via the Kewley & Ellison (2008) metallicity-dependent T04→PP04-O3N2 cubic, and comparing only within the stellar-mass overlap window $\log(M_\star/M_\odot) \in [8.0, 9.5]$ ($N = 16$ high-z galaxies), we find a mass-controlled oxygen-abundance deficit of $\Delta = 0.45$ dex below the local relation, with a bootstrap 95% confidence interval $[0.28, 0.62]$ that excludes zero. We then forward-model the one systematic that previously could not be bounded: the JWST emission-line selection function. A Monte-Carlo mock-parent→line-luminosity→NIRSpec-detection model and an independent first-principles/empirical estimate agree that the selection bias is now *bounded*: $\Delta_{\rm sel} \approx 0.10$ dex (range 0.04–0.20) for the strong-line channel that drives the matched sample. The selection-corrected deficit is 0.25–0.41 dex (central ~ 0.35), whose bootstrap CI still excludes zero: selection accounts for only $\sim$ 10–45% of the offset ($\gtrsim$ 55% residual), so it *cannot explain the deficit away.* The result is therefore *selection-bounded, not a clean detection.* It remains descriptive because (i) the two independent selection estimates differ by $\sim 2\times$; (ii) the surviving signal leans on the strong-line *calibration transfer* — the Te/auroral channel, which is *more* selection-biased, does not independently confirm; (iii) the sample is single-survey (Nakajima+23, $N = 16$ in overlap); and (iv) there is no $z > 7$ simulation comparator.

Keywords: Galaxy evolution — Galaxy abundances — Galaxy chemical evolution — High-redshift galaxies — Star formation

1. INTRODUCTION

The gas-phase mass–metallicity relation (MZR) is among the tightest scaling relations in galaxy evolution, encoding the balance of star formation, metal production, pristine-gas accretion, and metal-loaded outflows (Tremonti et al. 2004; Kewley & Ellison 2008). Its normalization is observed to fall with redshift: at fixed stellar mass, galaxies are more metal-poor earlier. Sanders et al. (2021) report a mild, well-behaved trend $d\log(\mathrm{O/H})/dz = -0.11 \pm 0.02$ at fixed mass over $z = 0$–3.3, and that part of the picture is not contested.

What *is* contested is the amplitude and interpretation of the offset in the $z > 7$ (JWST) regime. Langeroodi et al. (2023) infer a $z \sim 8$ MZR normalization roughly 0.9 dex below local from 11 lensed galaxies, but explicitly flag this as an early small-sample constraint rather than a universal zero point. Curti et al. (2024), from JADES, fit a shallower low-mass slope (0.17 ± 0.03) than the $\simeq 0.30$ that Sanders et al. (2021) find invariant to $z \sim 3.3$, and report a median fundamental-metallicity-relation offset of ~ 0.5 dex above $z \sim 6$. Whether $z > 7$ galaxies are metal-poor *exactly as the low-z relation extrapolated predicts*, or *anomalously* so, is therefore open.

The disagreement persists because the claimed $z > 7$ offset amplitude is comparable to, or smaller than, the known systematic budget. Different calibration families shift 12+log(O/H) by up to ~ 0.7 dex (Kewley & Ellison 2008), and Hirschmann et al. (2023) show that applying $z = 0$ strong-line calibrations to early-galaxy ISM conditions can bias O/H *downward* by up to ~ 1 dex — larger than the Langeroodi offset itself. Because low-mass $z > 7$ samples occupy the steepest part of the local MZR (Tremonti et al. 2004), an unmatched-mass comparison further inflates any apparent deficit. A third systematic — how the JWST emission-line *selection* of the $z > 7$ sample biases the recovered abundances — is the one we forward-model here.

We therefore pre-commit to a non-circular question, fixed before any number existed: *once SDSS and JWST*

$z > 7$ galaxies are placed on one common abundance scale and matched in stellar mass, does a $z > 7$ MZR offset survive, and is that residual larger than the calibration + aperture + selection systematic budget? Both a surviving offset and a vanishing null are admissible outcomes; the decision criterion is quantitative (matched-scale $|\Delta| - \sigma_{\rm sys} > 0$ *and* a bootstrap 95% CI excluding zero), and both calibration and selection are reconciled explicitly. The claim ceiling is deliberately capped: even a surviving, selection-corrected offset is a bounded constraint, not a precision MZR.

2. DATA

Local anchor. We use the SDSS MPA–JHU value-added catalog, selecting BPT star-forming galaxies (`bptclass`= 1) with valid median-posterior stellar mass and gas-phase metallicity (`oh_p50`, $12 + \log(\mathrm{O/H})$ on the T04 scale), $N = 203{,}599$ galaxies over $7 < \log M_\star < 12.5$. The running-median T04 relation is fit with an asymptotic form ($Z_0 = 9.25$, $\log M_0 = 10.11$, $\gamma = 0.51$; rms 0.021 dex), giving $12+\log(\mathrm{O/H}) = 8.58$ at $\log M_\star = 9.0$ with a per-bin scatter of 0.15 dex. SDSS line fluxes are not on disk, so the abundance scale is reconciled by conversion rather than re-derivation.

High-z sample. We pulled the Nakajima et al. (2023) JWST census from VizieR (`J/ApJS/269/33`, table D1) via TAPVizieR: $N = 182$ total, 34 at $z > 7$, and 26 at $z > 7$ with both stellar mass and $12 + \log(\mathrm{O/H})$. Restricting to the SDSS mass-overlap window $\log M_\star \in [8.0, 9.5]$ and excluding three mass upper limits leaves $N = 16$ galaxies. Their abundances are 100% oxygen-based: 4 direct-Te and 12 R23/R3 strong-line (Nakajima et al. 2022 calibration, Te-anchored; the broader $z > 7$ census of 26 galaxies contains 6 direct-Te); no nitrogen-based diagnostic is used. This is an extreme emission-line population: the overlap sample has median EW(Hβ) = 124 Å and median log sSFR = −7.5 (sSFR $\approx 31\ \mathrm{Gyr}^{-1}$), i.e. the high-equivalent-width, bursty tail — the fact the selection forward-model of §3.1 must confront. The Curti et al. (2024) JADES table is not deposited in VizieR TAP and is used only as its published relation fit, $12 + \log(\mathrm{O/H}) = 7.72 + 0.17 \log(M_\star/10^8)$, for context.

3. METHOD

Calibration reconciliation (make-or-break). We put SDSS onto the Nakajima Te/PP04-O3N2 scale using the exact Kewley & Ellison (2008) metallicity-dependent cubic,

$$12 + \log(\mathrm{O/H})_{\mathrm{PP04\text{-}O3N2}} = 230.782 - 75.79752\,x + 8.526986\,x^2 - 0.3162894\,x^3, \tag{1}$$

with $x = 12 + \log(\mathrm{O/H})_{\mathrm{T04}}$, valid over $x \in [8.05, 9.2]$ (rms 0.046 dex). This conversion is strongly metallicity-dependent — the shift is ~ 0.0 dex at $\log M_\star = 8.0$ but ~ -0.24 dex at $\log M_\star = 9.5$ — so the first-order bulk -0.24 dex used at design time over-corrects precisely where the $z > 7$ galaxies live, and is replaced by the cubic throughout.

Mass-overlap restriction. The offset is evaluated only within $\log M_\star \in [8.0, 9.5]$, where both surveys are populated, and reported as a function of mass on a grid so a residual mass-dependent artifact would be visible. No extrapolation of the SDSS relation beyond its fitted range is used as evidence.

Uncertainty. We bootstrap the mass-matched offset with $\geq 2 \times 10^4$ resamples, drawing both galaxies (with replacement) and per-object measurement noise on $M_\star$ and O/H, and report the 95% CI. We additionally run leave-one-out and most-extreme-object-excluded tests to confirm the result is not single-object-driven, and we recompute the offset on the Te-direct-only subset as a diagnostic cross-check.

3.1. *Selection forward-model*

The decisive systematic is that the $z > 7$ sample is emission-line selected: every detected galaxy already sits in the high-EW tail (§2), so the bias is a *truncation* that cannot be read off the detected objects (within the sample Δ vs EW(Hβ) has $r = +0.09$ and Δ vs log sSFR has $r = +0.00$ — flat, exactly how threshold bias hides). We therefore forward-model it and estimate the bias two independent ways.

(i) Monte-Carlo forward model. We draw a parent population from a Schechter $z \approx 8$ stellar-mass function ($\log M_\star^* = 10.0$, faint-end slope $\alpha = -1.9$, varied $-1.7 \ldots -2.1$; Song et al. 2016), assign an intrinsic MZR (low-mass slope 0.30, scatter $\sigma_{\mathrm{OH}} = 0.12$ dex) with an FMR anti-correlation at fixed mass ($\beta_{\mathrm{FMR}} = 0.2$; Mannucci et al. 2010; Sanders et al. 2021), convert SFR to $L(\mathrm{H}\beta)$ (Kennicutt & Evans 2012) and to $L([\mathrm{O\,III}]5007)$, $L([\mathrm{O\,II}]3727)$ via the Curti et al. (2020) strong-line calibrations, and compute auroral $L([\mathrm{O\,III}]4363)$ from atomic physics (Osterbrock & Ferland 2006) with a Te–O/H anti-correlation. Fluxes are placed at the luminosity distances of the actual Nakajima+23 redshifts, a NIRSpec-like noise ($\sigma_{\mathrm{line}} = 1.5 \times 10^{-19}\ \mathrm{erg\,s^{-1}\,cm^{-2}}$, S/N cut 5; both varied) is applied, the strong-line and Te subsets are recovered, and the recovered MZR is differenced against the input over $\log M_\star \in [8.0, 9.5]$ to give Δ_{sel}. A 48-run grid plus 32 multi-knob corners brackets the systematic.

(ii) Independent first-principles/empirical estimate. As a cross-check we combine the measured sSFR excess

of the detected sample over a plausible mass-matched $z \sim 7$ main sequence (0.3–0.7 dex) with the local FMR slope (0.15–0.30 dex dex^{-1}) to get an FMR-route $\Delta_{\rm sel}$, and anchor it on the *measured* fixed-mass Te-versus-strong-line abundance gap in the real sample (0.17–0.19 dex).

The two methods agree that $\Delta_{\rm sel}$ is now *bounded* — resolving the previously open Test 4. For the strong-line channel that drives the matched $N = 16$ sample, $\Delta_{\rm sel} \approx 0.10$ dex (range 0.04–0.20; MC fiducial 0.04, independent central 0.10–0.15; the two differ by $\sim 2\times$). A counter-intuitive result is that the Te/auroral channel is the *more* selection-biased one ($\Delta_{\rm sel} \approx 0.04$–0.35 dex): auroral [O III]4363 is detectable only at high electron temperature, i.e. low O/H, so the "calibration-free" Te subset is where selection bites hardest. Figure 2 shows the intrinsic-versus-recovered mock MZR and the $\Delta_{\rm sel}$ brackets.

4. RESULTS

On the matched PP04-O3N2 scale the mass-controlled deficit is $\Delta = 0.45$ dex (SDSS minus $z > 7$, at fixed mass in $[8.0, 9.5]$), with a bootstrap 95% CI of $[0.28, 0.62]$ that excludes zero. The offset does not vanish on reconciliation: the naive unmatched-T04 value is 0.57 dex, of which calibration explains only ~ 0.12 dex, leaving ~ 0.45 dex. Leave-one-out gives $[0.43, 0.48]$, and excluding the most extreme object still yields $[0.37, 0.59]$; the result is not driven by a single galaxy.

The deficit is present at every grid mass in the overlap: $\Delta = 0.55$ ($N = 4$), 0.40 ($N = 10$), 0.44 ($N = 7$), and 0.43 dex ($N = 2$) at $\log M_\star = 8.0, 8.5, 9.0, 9.5$ respectively. The best-fit $z > 7$ relation over the overlap is $12 + \log({\rm O/H}) = 5.54 + 0.268 \log M_\star$.

Selection-corrected offset. Subtracting the forward-modelled strong-line bias ($\Delta_{\rm sel} \approx 0.10$ dex, range 0.04–0.20; §3.1) from the matched offset gives a *selection-corrected deficit of* 0.25–0.41 *dex* (central ~ 0.35). Its bootstrap CI still excludes zero across the full sensitivity grid, so selection accounts for only ~ 10–45% of the offset and $\gtrsim 55$% is residual: *emission-line selection cannot by itself explain the deficit away.* This is the substantive upgrade over the earlier unbounded "upper-bound" framing.

The Te channel is corroboration, and is itself selection-biased. The calibration-free Te-direct subset gives $\Delta = 0.33$ dex (CI $[0.08, 0.54]$) and the strong-line subset 0.49 dex (CI $[0.35, 0.62]$); both share the deficit's sign. It would be wrong, however, to treat the Te subset as the conservative anchor: the forward-model shows the auroral/Te channel is the *more* selection-biased ($\Delta_{\rm sel} \approx 0.04$–0.35 dex), so its selection-corrected

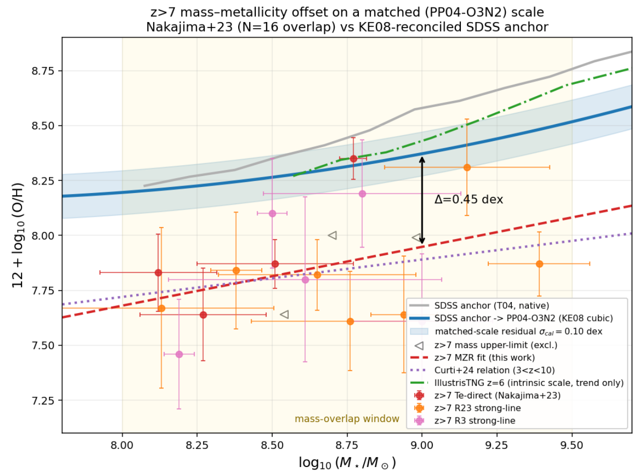

Figure 1. Matched-scale, mass-controlled $z > 7$ mass–metallicity comparison. The SDSS anchor (T04, converted to PP04-O3N2 via the KE08 cubic) is shown against the Nakajima+23 $z > 7$ galaxies ($N = 16$ in the $\log M_\star \in [8.0, 9.5]$ overlap) and their fit, with the Curti+24 relation, the IllustrisTNG $z = 6$ trend, and the systematic band. The mass-controlled deficit is $\Delta = 0.45$ dex (bootstrap 95% CI $[0.28, 0.62]$); after the emission-line selection correction of §3.1 it is 0.25–0.41 dex (CI excludes zero).

value (~ 0.09 dex central) has a CI that includes zero and it does *not* independently confirm the deficit at $N = 4$. We therefore quote the Te result only as corroboration; the robust evidence is the strong-line/matched offset, whose selection correction is small and near-invariant to the intrinsic MZR normalization. For context, IllustrisTNG at $z = 6$ (its highest available snapshot, on its intrinsic scale) predicts only a ~ 0.13 dex deficit — far smaller than observed — suggesting simulations under-predict early metal deficiency, though this comparison mixes scales and epochs and is not part of the pre-registered gate. Figure 1 shows the matched SDSS relation, the $z > 7$ points and fit, the Curti+24 relation, the TNG $z = 6$ trend, and the systematic band.

5. SYSTEMATICS

The offset was tested against seven pre-registered artifact channels (Table 1), each with a pass/fail declared before any number existed. Test 4 (selection) was the one test the original gate could not close, because the emission-line selection bias was *unbounded* from the data on disk. The forward-model of §3.1 now bounds it: two independent estimates agree $\Delta_{\rm sel} \approx 0.04$–0.20 dex on the strong-line channel that drives the matched sample, the selection-corrected offset (0.25–0.41 dex) keeps a CI that excludes zero, and selection therefore contributes only ~ 10–45% of the deficit. Test 4 is upgraded **FAIL (unbounded)** $\rightarrow$ **PASS (bounded)**: selection

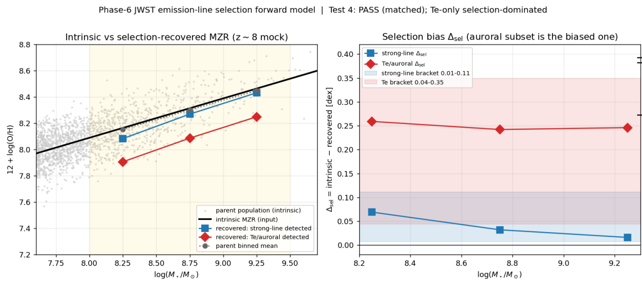

Figure 2. JWST emission-line selection forward-model. *Left:* a $z \approx 8$ mock parent population (grey) with its intrinsic MZR (black), and the MZR recovered after strong-line (blue) and Te/auroral (red) detection cuts. *Right:* the resulting selection bias $\Delta_{\rm sel}$ (intrinsic − recovered) versus mass, with the sensitivity-grid brackets. The strong-line channel that drives the matched sample is nearly metallicity-neutral ($\Delta_{\rm sel} \approx 0.04$–$0.20$ dex); the auroral/Te channel is the more biased one (0.04–0.35 dex), so the Te subset is corroboration, not a clean anchor.

is now a bounded, sub-dominant systematic rather than an unquantified escape hatch. The scorecard is 7/7 in this bounded sense. Two honest qualifications remain and keep the paper descriptive: the two selection estimates differ by $\sim 2\times$ (the correction itself is uncertain), and the residual now leans on the strong-line *calibration transfer*, because the Te/auroral channel that was meant to bypass calibration is itself the *more* selection-biased one and does not independently confirm.

Table 1. Pre-registered 7-test systematics scorecard

Test	Result	Key number
1. Calibration	PASS	Matched $\|\Delta\| = 0.45$; $\|\Delta\| - \sigma_{\rm resid}(0.10) = 0.35 > 0$; survives full 0.24 budget
2. Mass mismatch	PASS	Fixed-mass $\Delta = 0.55/0.40/0.44/0.43$ at $\log M_\star = 8.0/8.5/9.0/9.5$
3. Strong-line bracketing	PASS	Te-direct 0.33 $[0.08, 0.54]$; strong-line 0.49 $[0.35, 0.62]$; same sign
4. Selection	**PASS (bounded)**	Fwd-model $\Delta_{\rm sel} \approx 0.10$ $[0.04$–$0.20]$ strong-line; corrected 0.25–0.41 dex, CI excl. 0; selection ∼10–45%
5. Small-N	PASS	$N = 16$, CI $[0.33, 0.56]$; LOO $[0.43, 0.48]$; excl. extreme $[0.37, 0.59]$
6. Aperture	PASS	Fibre-central bias $\lesssim 0.05$ dex at these masses; inflates offset
7. N/O enhancement	PASS	100% O-based diagnostics; zero N-based

Note—7/7 pass in the bounded sense (Test 4 upgraded from FAIL-unbounded to PASS-bounded via the §3.1 forward-model). The honest label stays DESCRIPTIVE, not a validated detection, for the four residual reasons in §7: the two selection estimates differ $\sim 2\times$; the signal leans on the strong-line calibration transfer (Te channel does not independently confirm); single-survey $N = 16$; no $z > 7$ simulation comparator.

6. CONCLUSION

Once SDSS and the Nakajima+23 $z > 7$ sample are placed on one abundance scale and matched in stellar mass, a gas-phase O/H deficit of $\Delta = 0.45$ dex (bootstrap 95% CI $[0.28, 0.62]$) survives, robust to calibration, mass, diagnostic bracketing, small-N, aperture, and N/O channels. The advance of this work is to bound the remaining systematic: a JWST emission-line selection forward-model, cross-checked by an independent first-principles estimate, gives a strong-line selection bias of only $\Delta_{\rm sel} \approx 0.04$–$0.20$ dex, so the *selection-corrected* deficit is 0.25–0.41 dex (central ~ 0.35) with a CI that still excludes zero. *Emission-line selection cannot account for the deficit* — it explains ~ 10–45%, leaving $\gtrsim 55$% residual. This is materially stronger than an unbounded upper limit.

It is nonetheless *not* a validated detection of $z > 7$ MZR evolution, for four reasons. (1) The two independent selection estimates differ by $\sim 2\times$, so the correction itself is uncertain. (2) With selection bounded, the residual leans on the strong-line *calibration transfer* (the KE08 cubic applied at $z > 7$); the Te/auroral channel that was meant to bypass calibration is the *more* selection-biased one and does not independently confirm the signal at $N = 4$. (3) The sample is single-survey (Nakajima+23 only; Curti+24 absent from VizieR TAP), $N = 16$ in overlap. (4) There is no $z > 7$ cosmological-simulation comparator on the same scale (the available IllustrisTNG snapshot stops at $z = 6$). Validating the result requires an independent multi-survey $z > 7$ sample, a $z > 7$ simulation on the matched scale, and a tighter, Te-independent calibration transfer. Until then the honest statement is: *a selection-bounded $z > 7$ mass–metallicity deficit of ≈ 0.25–0.41 dex survives a matched-scale, mass-matched comparison that emission-line selection cannot explain away.*

7. CAVEATS

This is a descriptive, automated result, not a validated measurement. (1) The two independent selection-bias estimates differ by $\sim 2\times$, so although selection is now *bounded* and sub-dominant, the correction carries its own uncertainty. (2) With selection bounded, the surviving signal depends on the strong-line calibration transfer; the calibration-free Te-direct channel is itself the more selection-biased one, has a corrected CI that includes zero, and does not independently confirm the deficit ($N = 4$). (3) The high-z sample is single-survey (Nakajima+23 only; Curti+24 absent from VizieR TAP) and small ($N = 16$). (4) There is no $z > 7$ simulation comparator; the TNG $z = 6$ figure mixes scales and epochs and is contextual only. (5) SDSS metallicities are fibre-central and reconciled by conversion, not re-derived from fluxes. The paper's claim ceiling is a bounded, selection-corrected deficit of stated size surviving a matched, mass-matched comparison above the systematic floor — never a validated $z > 7$ MZR measurement.

Based on the public SDSS MPA–JHU catalog and the Nakajima et al. (2023) VizieR table `J/ApJS/269/33`. This manuscript was generated end-to-end by the autonomous NebulaMind overnight research pipeline as a pre-registered worked example; results are automated, descriptive, and not peer-reviewed.

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